EXPERIMENT 10

Mobile Usage Statistics using Data Visualization

AIM:

The aim of this experiment is to create data visualizations such as pie charts and bar graphs to represent mobile usage statistics using JavaScript.

OBJECTIVE:

To analyze and visualize the time spent on different mobile applications using graphical representation.

PROCEDURE:

Step 1: Create HTML File

Create an HTML file (index.html) to design the structure of the webpage. Include the Google Charts library in the head section and create space for displaying charts using elements.

Step 2: Load Google Charts

Load the Google Charts library using the loader script and initialize it by specifying the required chart package.

Step 3: Prepare the Data

Define the data representing time spent on various mobile applications such as Instagram, WhatsApp, YouTube, Games, and Others.

Step 4: Create Pie Chart

Use the PieChart function to visualize the distribution of time spent on different mobile applications.

Step 5: Create Bar Chart

Use the BarChart function to compare the time spent across different applications in a graphical format.

Step 6: Apply Styling

Use CSS to center-align the charts and improve the overall appearance of the webpage.

Step 7: Execute the Program

Save the file and open it in a web browser to view the generated charts.

SOURCE CODE:

```
<!DOCTYPE html>
<html>
<head>
  <title>Mobile Usage Statistics</title>

  <script src="https://www.gstatic.com/charts/loader.js"></script>

  <style>
    body {
      font-family: Arial, sans-serif;
      text-align: center;
      margin: 0;
      padding: 0;
    }

    .container {
      display: flex;
      flex-direction: column;
      align-items: center;   /* center horizontally */
      justify-content: center;
```

```
    margin-top: 30px;
  }

  .chart-box {
    margin: 20px;
  }
</style>

<script>
  google.charts.load('current', {'packages':['corechart']});
  google.charts.setOnLoadCallback(drawCharts);

  function drawCharts() {

    var data = google.visualization.arrayToDataTable([
      ['App', 'Hours per Day'],
      ['Instagram', 3],
      ['WhatsApp', 2],
      ['YouTube', 4],
      ['Games', 2],
      ['Others', 1]
    ]);

    var pieChart = new
google.visualization.PieChart(document.getElementById('piechart'));
    pieChart.draw(data, {title: 'Mobile App Usage Distribution'});
```

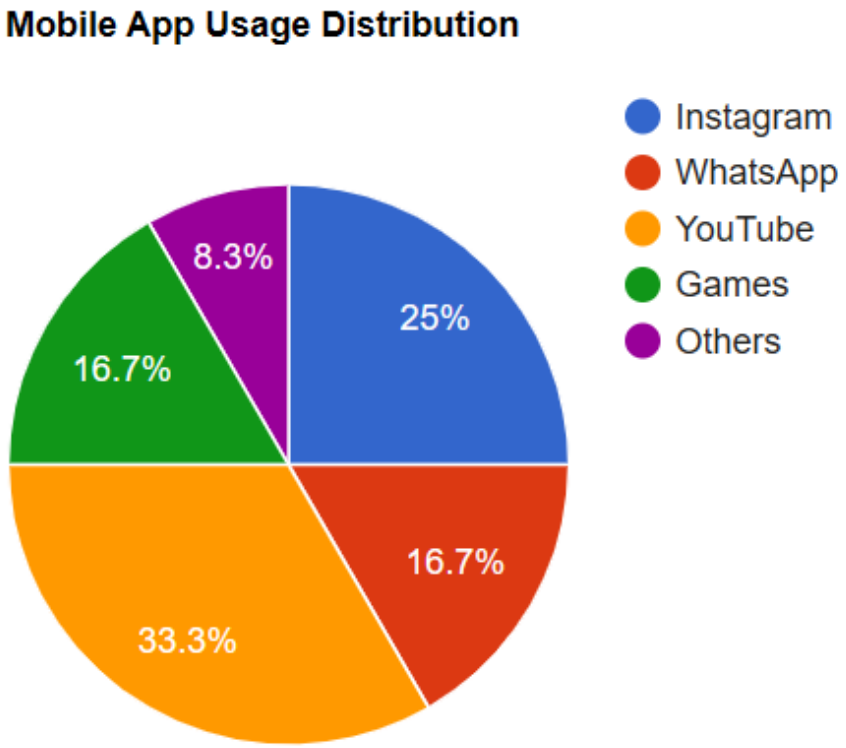
```
    var barChart = new
google.visualization.BarChart(document.getElementById('barchart'));
    barChart.draw(data, {
      title: 'Time Spent on Apps',
      hAxis: {title: 'Hours'},
      vAxis: {title: 'Apps'}
    });
  }
 </script>
</head>

<body>

 <h1>Mobile Usage Statistics</h1>

 <div class="container">
  <div id="piechart" class="chart-box"
style="width:500px;height:400px;"></div>
  <div id="barchart" class="chart-box"
style="width:500px;height:400px;"></div>
 </div>

</body>
</html>
```

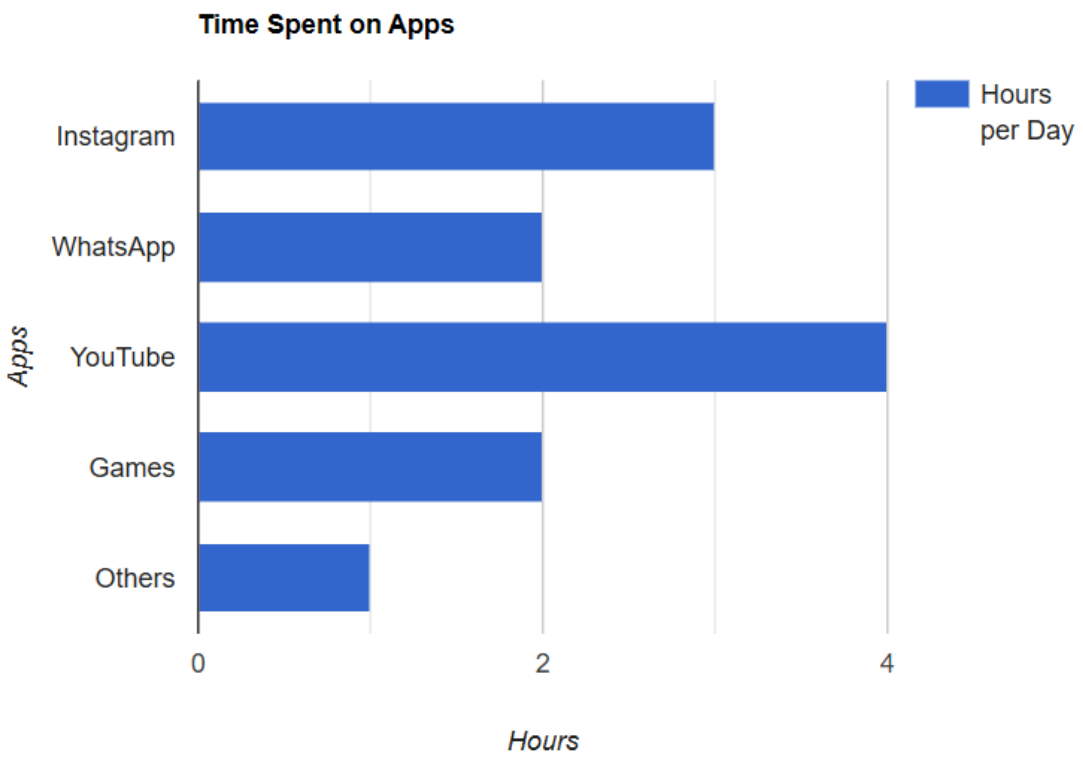
OUTPUT:

The output displays:

- A pie chart representing the distribution of mobile usage across different applications.
- A bar chart showing the comparison of time spent on each application.

Mobile Usage Statistics





RESULT:

Thus, the mobile usage statistics were successfully visualized using pie charts and bar graphs with the help of Google Charts.